

Low Energy Electron Diffraction (low energy electron diffraction (LEED))

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1 Introduction

LEED is a powerful and standard technique for investigating the surface structure of crystalline materials by the diffraction of low energy electrons. The technique was first used by Davisson and Germer in 1927 on a Nickel crystal.^[1] Since then, LEED has become a widely used technique in surface science and materials research.^[2]

The basic principle of LEED is to direct a beam of low-energy electrons onto a crystalline surface and to analyze the resulting diffraction pattern. The diffraction pattern is formed by the interference of the scattered electrons from the surface atoms, which leads to a characteristic pattern of bright spots on a fluorescent screen. The positions and intensities of these spots provide information about the surface structure, such as the lattice constant, surface reconstruction, and orientation of the surface.^[2]

The LEED patterns formed by the scattered electrons show the reciprocal lattice of the surface and adsorbed structures. The reciprocal lattice vectors $\mathbf{a}_j^*$ and the real space lattice vectors $\mathbf{a}_i$ are related by the following equation:

$$\mathbf{a}_i \cdot \mathbf{a}_j^* = 2\pi\delta_{ij}, \quad (1)$$

where δ_{ij} is the Kronecker delta, which is 1 if $i = j$ and 0 otherwise.

In this report, LEED is used to investigate the surface structure of Au(111) and the adsorption of perylenetetracarboxylic dianhydride (PTCDA) on this surface. The Au(111) surface is a well-studied model system in surface science due to its simple structure and well-defined surface properties.^[3,4]

The Au(111) surface has a hexagonal close-packed structure with a lattice constant of approximately 2.88 Å.^[4] The surface is known to undergo a $(22\times\sqrt{3})$ reconstruction.^[3]

The LEED experiments in this report aim to determine the surface lattice constant, the bulk lattice constant, and the reconstruction of the Au(111) surface. Furthermore, the adsorption of PTCDA on the Au(111) surface is investigated by measuring the LEED pattern of PTCDA on the Au(111) surface.

2 Experimental

2.1 Experimental Setup

The LEED measurements were conducted in an ultra-high vacuum (UHV) chamber with a base pressure below $5 \cdot 10^{-10}$ mbar. The chamber was equipped with a turbomolecular pump and an ion getter pump. The sample was mounted on a sample holder that allowed for precise rotation and translation in all directions. As sample, a single crystal of Au(111) was used, which was chosen for its well-known surface structure. The sample holder was equipped with a resistive heating element to facilitate annealing procedures and a thermocouple for temperature measurement.

Furthermore, the chamber was equipped with a LEED optics system, which included an electron gun, a hemispherical fluorescent screen and a camera for recording the diffraction patterns.

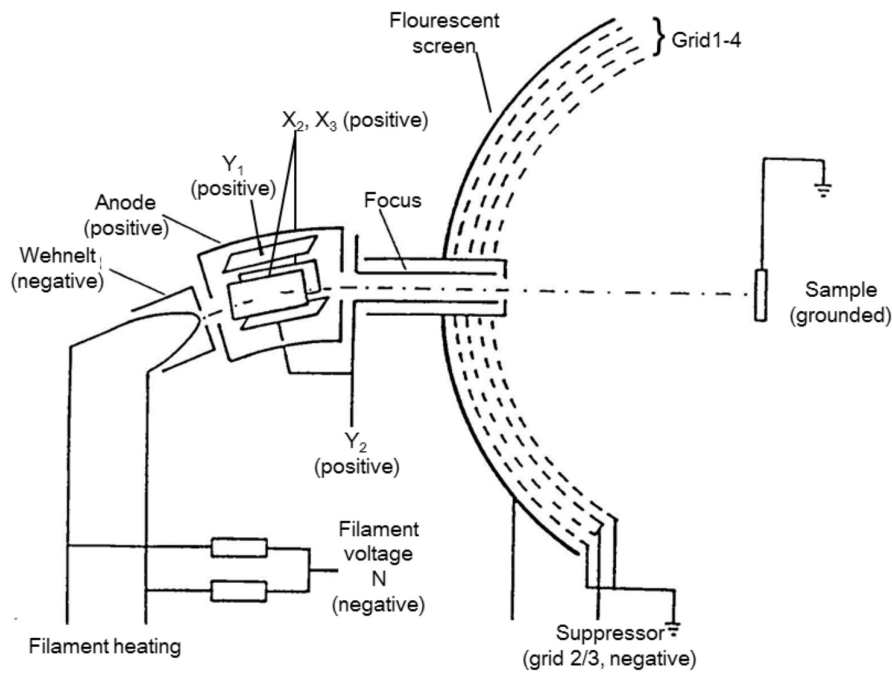


Figure 1: Schematic (cross section) of the LEED experimental setup including the individual voltages for the electron gun and the fluorescent screen. Taken from reference [5].

The LEED optics has several components to generate and focus the electron beam onto

the sample surface. The electron gun generates a beam of low-energy electrons, ranging from 50 to 200 eV, which is then focused by a series of lenses onto the sample surface. The electron beam is then scattered by the surface atoms, producing a diffraction pattern that is recorded by the fluorescent screen and camera. The parameters of the LEED optics were optimized, using the provided interface on the computer, to achieve a high resolution and a low background noise.

2.2 Sample Preparation

The Au(111) single crystal surface was prepared in accordance with standard UHV procedures. The Au(111) single crystal was subjected to an initial cleaning process involving the sputtering with Ar^+ ions with an energy of 1 keV for a duration of 30 minutes. This was followed by a subsequent annealing procedure at a temperature of 750 K for a period of 60 minutes. The cleaning procedure was executed at the onset of the experiment to guarantee a clean Au(111) surface for the LEED measurements. Subsequently, the sample was then cooled down to room temperature, thus preparing it for the LEED measurements.

After the LEED measurements on the Au(111) surface, PTCDA was evaporated onto the sample. Therefore, the evaporator was heated to a specific temperature and the sample was then rotated towards the evaporator to facilitate the deposition process.

2.3 LEED Measurements

LEED measurements were performed using LEED system explained before. The electron gun was operated at various energies between 25-200 eV. The filament heating was set to a constant current of 3.1 A. The electron beam was focused onto the sample surface, and the resulting diffraction pattern was depicted on the fluorescent screen. The diffraction patterns were captured using a camera, which allowed for high-resolution imaging of the diffraction spots.

With the optimized LEED setup, several diffraction patterns were recorded at different electron energies and angles. First, the pure Au(111) surface was measured to obtain a reference pattern. Subsequently, the sample was rotated in steps of 5° to investigate the influence of the surface orientation on the diffraction pattern.

Afterwards, the diffraction pattern of the Au(111) surface was recorded at different electron energies, ranging from 80 eV to 200 eV. The diffraction patterns were analyzed to determine the spot distances and the lattice constant of the Au(111) surface. In addition, the reconstruction of the Au(111) surface was investigated by measuring the diffraction pattern. The LEED patterns were analyzed to determine the Au(111) surface reconstruction.

Finally, PTCDA was deposited on the Au(111) surface by evaporation. The LEED pattern of PTCDA on the Au(111) surface was recorded at an electron energy of 23.4 eV to investigate the epitaxial growth of PTCDA on the Au(111) surface.

3 Results

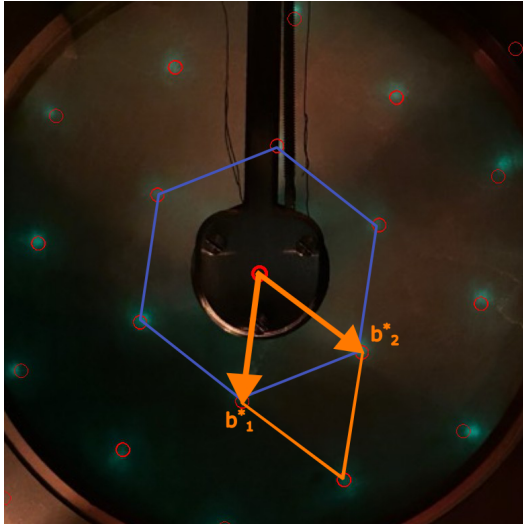
The results of the LEED measurements are presented in the following sections. The first section discusses the orientation of the Au(111) surface in the diffraction pattern, followed by a section on the determination of the distance between the sample and the LEED screen. Subsequently, the analysis of the LEED pattern of the Au(111) surface is presented, including the determination of the lattice constant and the reconstruction of the Au(111) surface. Finally, the last section discusses the LEED pattern of PTCDA on the Au(111) surface.

3.1 Orientation of the surface

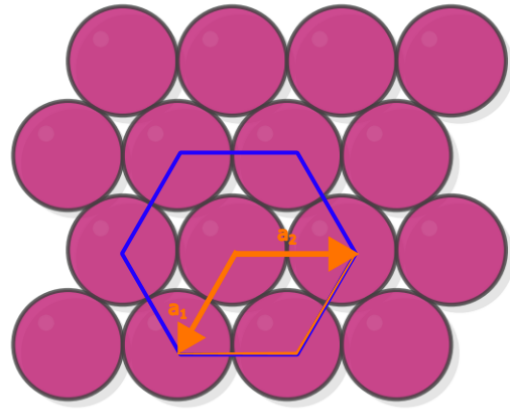
The orientation of the Au(111) surface in the diffraction pattern in comparison to a hard-sphere model was determined using the LEED pattern obtained at an electron energy of 179.3 eV. The diffraction spots were analyzed and compared to the expected positions of the diffraction spots for a perfect Au(111) surface. Figure 2 shows the surface orientation of the Au(111) crystal structure in comparison to the hard-sphere model.

The diffraction spots in the LEED pattern are arranged in a hexagonal pattern, which is consistent with the hard-sphere model of the Au(111) surface. Furthermore, it can be observed that the angle between the reciprocal lattice vectors of the Au(111) surface is approximately 60° , while the real space lattice vectors are at an angle of 120° . The reason for this difference is that the following relation holds for the reciprocal lattice vectors $\mathbf{b}$ and the real space lattice vectors $\mathbf{a}$:

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij} \text{ with } \mathbf{a}_i \cdot \mathbf{b}_j = |\mathbf{a}_i||\mathbf{b}_j| \cos(\psi). \quad (2)$$



(a) LEED pattern of Au(111) at an energy of 179.3 eV



(b) Hard-sphere model of the Au(111) surface

Figure 2: Comparison of the LEED pattern of the Au(111) surface at an electron energy of 179.3 eV with a hard-sphere model of the Au(111) surface. The red circles indicate the positions of the diffraction spots in the LEED pattern. The blue hexagon shows the symmetry of the Au(111) surface. The orange vectors indicate the lattice vectors of the Au(111) surface. The orange lines show the unit cell.

3.2 Determination of the sample distance to the LEED screen

The spot distances in the LEED pattern were measured to determine the distance r between the Au(111) sample and the detector, which is equivalent to the screen radius. The distances between the diffraction spots were extracted from the LEED patterns in Figure 3 recorded at the same electron energy of 119.6 eV but at different angles. The distances were measured in pixels.

The distances between the diffraction spots were measured in pixels and are summarized in Table 1. The distances were measured from the center of the diffraction pattern to the center of the diffraction spots, but only the projection in the radial direction was considered.

Table 1: Measured distances between the diffraction spots in pixels for different angles.

angle θ ($^\circ$)	distance Δx [px]
0	0
5	132
10	192
15	328

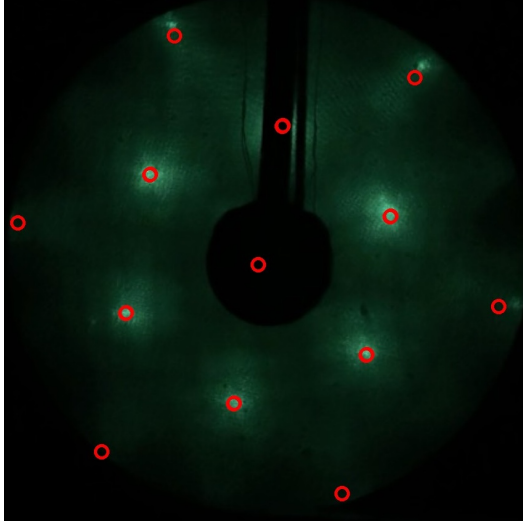
With the measured distances of the diffraction spots, the distance r between the Au(111) sample and the LEED screen can be determined using the formula:

$$\sin(2\theta) = \frac{\Delta x}{r}, \quad (3)$$

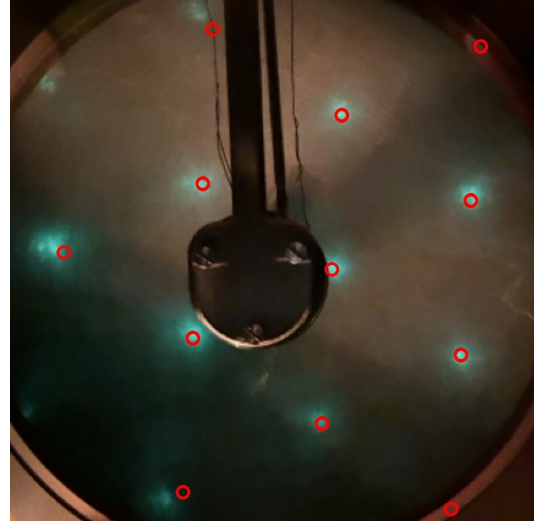
where Δx is the distance between the diffraction spots in pixels and θ is the angle of the rotation.

Linearisation of equation 3 yields the distance r by plotting $\sin(2\theta)$ against Δx . The slope of the linear fit to the data points gives the value of $1/r$. The resulting plot is shown in Figure 4.

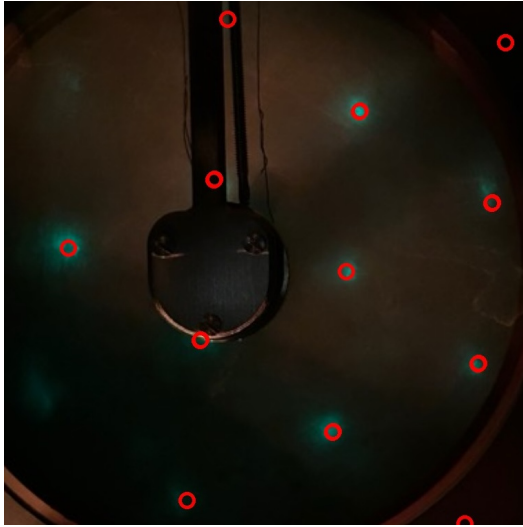
The result of the linear fit is shown in Figure 4. The slope of the linear fit is 0.001565 ± 0.000164 , which corresponds to a distance of $r = 638.8 \pm 66.8$ px between the Au(111) sample and the LEED screen. The intercept of the linear fit is -0.0001 ± 0.0002 , which is more or less consistent with the expected value of zero, as the distance should be zero at an angle of zero.



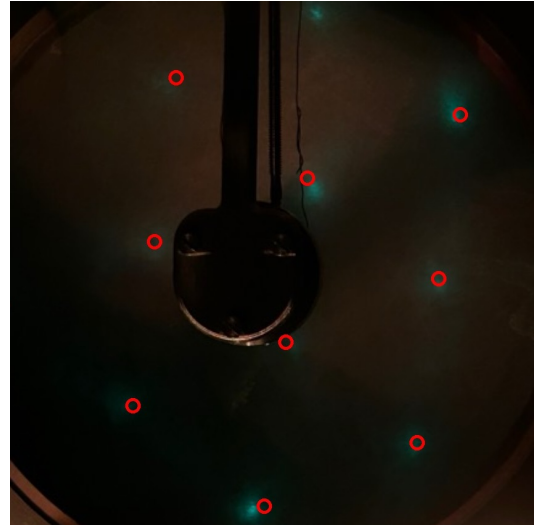
(a) 0° rotation



(b) 5° rotation



(c) 10° rotation



(d) 15° rotation

Figure 3: LEED patterns of Au(111) surface at an electron energy of 119.6 eV and with different angles. The red circles indicate the positions of the diffraction spots.

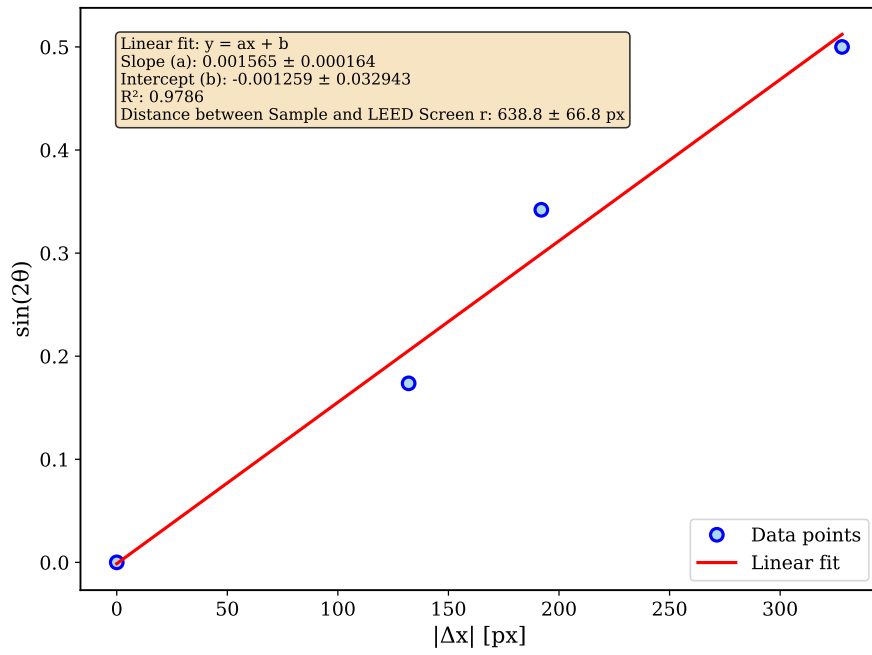


Figure 4: Plot of $\sin(2\theta)$ against the measured distances Δx between the diffraction spots in pixels. The slope of the linear fit gives the value of $1/r$, where r is the distance between the Au(111) sample and the detector.

3.3 Determination of the lattice constant of the Au(111) surface

The lattice constant of the Au(111) surface can be determined from the positions of the diffraction spots d in the LEED patterns and the distance between the sample and the LEED screen r , which was determined from the linear fit in Figure 4. The distance between the spots d is related to the lattice constant by the formula:

$$\mathbf{a}^* = |\mathbf{a}^*| = \frac{d}{r} |\mathbf{k}_0|, \quad (4)$$

where $\mathbf{a}^*$ is the reciprocal lattice vector, d is the distance between the diffraction spots in pixels, r is the distance between the sample and the LEED screen in pixels and $|\mathbf{k}_0|$ is the wave vector of the electrons, which can be calculated from the electron energy E using the formula:

$$|\mathbf{k}_0| = \frac{2\pi}{\lambda_e} = \frac{2\pi}{\sqrt{\frac{h^2}{2m_e E}}}, \quad (5)$$

where h is the Planck constant, m_e is the electron mass and E is the electron energy in eV. The wavelength λ_e of the electrons can be calculated from the electron energy using the formula:

$$\lambda_e = \frac{h}{\sqrt{2m_e E}}. \quad (6)$$

The wave vector $|\mathbf{k}_0|$ can be calculated for each electron energy used in the LEED measurements.

The LEED patterns of the Au(111) surface at different electron energies are shown in Figure 5. The distances between the diffraction spots were measured in pixels and are summarized in ???. The distances were measured from the center of the diffraction pattern to the center of the diffraction spots. For each electron energy, the distances were measured for six different diffraction spots and the average distance was calculated. The average distance is shown in the last row of ???.

With the measured distances of the diffraction spots d , the lattice constant in reciprocal

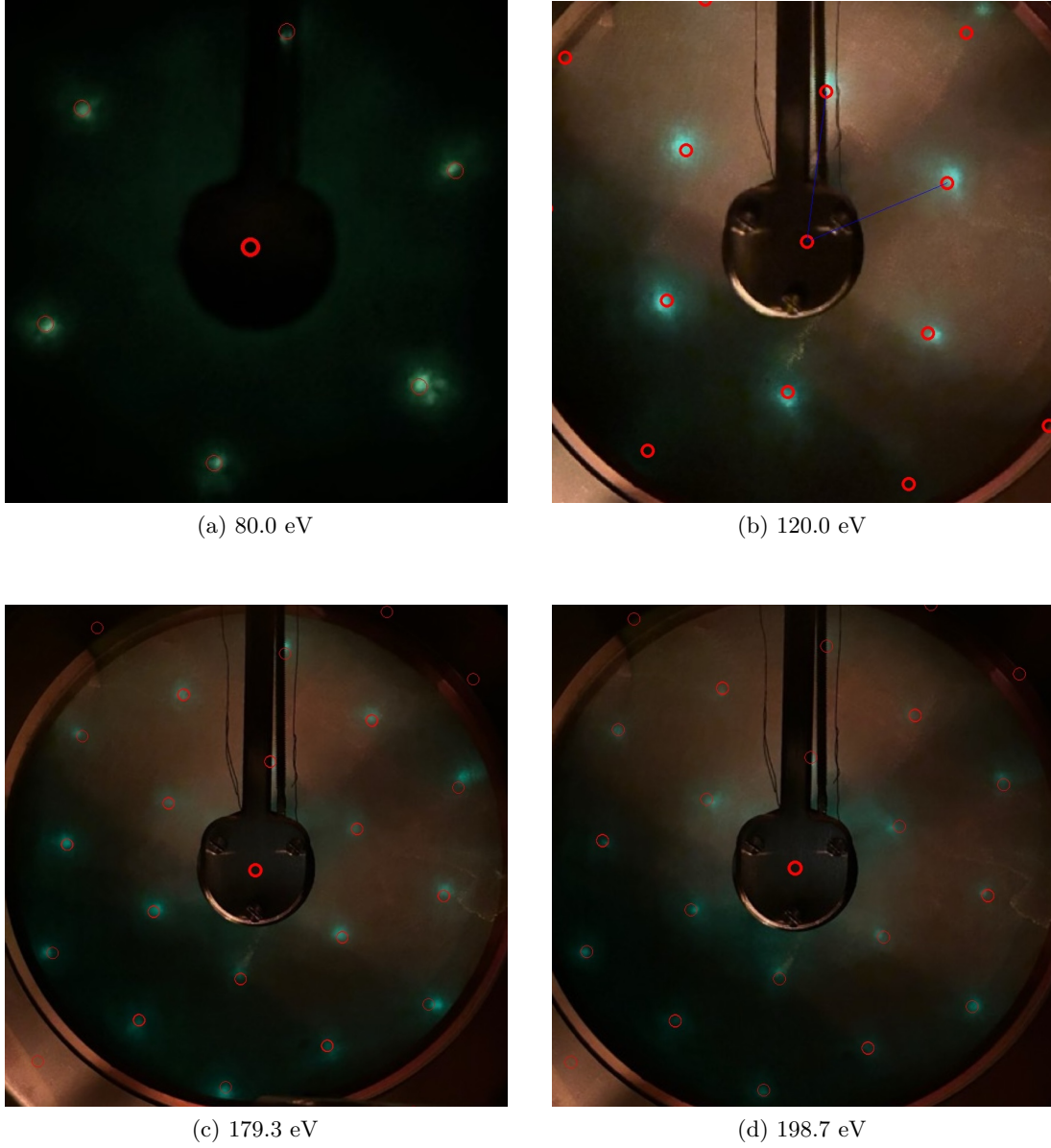


Figure 5: LEED patterns of Au(111) surface at different electron energies showing the characteristic diffraction pattern. The red circles indicate the positions of the diffraction spots.

space a^* of the Au(111) surface can be determined using Equation 4. Using the reciprocal lattice vector $\mathbf{a}^*$, the surface lattice constant $\mathbf{a}$ in real space can be calculated using the formula:

$$\mathbf{a} = |\mathbf{a}| = \frac{4\pi}{\sqrt{3}|\mathbf{a}^*|}. \quad (7)$$

The bulk lattice constant $\mathbf{a}_b$ can be calculated from the surface lattice constant $\mathbf{a}$ using the formula:

$$\mathbf{a}_b = |\mathbf{a}_b| = \sqrt{2} \cdot |\mathbf{a}|. \quad (8)$$

Table 2 summarizes the measured distances between the diffraction spots d in pixels, the calculated wave vector $|\mathbf{k}_0|$ in $\text{\AA}^{-1}$, the calculated reciprocal lattice constant $\mathbf{a}^*$ in $\text{\AA}^{-1}$, the calculated surface lattice constant $\mathbf{a}$ in $\text{\AA}$, and the calculated bulk lattice constant $\mathbf{a}_b$ in $\text{\AA}$ for the Au(111) surface at different electron energies. Furthermore, it is noteworthy that the obtained values for the electron energy of 80.0 eV and 119.6 eV are very similar larger than the values for the electron energies of 179.3 eV and 198.7 eV. This indicates that there is a systematic error in the measurements.

The final values for the reciprocal lattice constant, the surface lattice constant, and the bulk lattice constant of the Au(111) surface, averaged over all measured electron energies, are:

$$\begin{aligned} \mathbf{a}^* &= 2.29 \pm 0.24 \text{ \AA}^{-1} \\ \mathbf{a} &= 3.17 \pm 0.33 \text{ \AA} \\ \mathbf{a}_b &= 4.49 \pm 0.47 \text{ \AA} \end{aligned}$$

These values are in agreement with the literature values for the Au(111) surface, which are 2.8810 $\text{\AA}$ for the surface lattice constant and 4.0743 $\text{\AA}$ for the bulk lattice constant.^[4]

Table 2: Measured LEED spot distances and calculated lattice parameters for Au(111) surface at different electron energies.

E [eV]	λ_e [Å]	$\mathbf{k}_0$ [Å ⁻¹]	d [pixel]	$\mathbf{a}^*$ [Å ⁻¹]	$\mathbf{a}$ [Å]	$\mathbf{a}_b$ [Å]
80.0	1.371	4.583	307.0	2.20 ± 0.23	3.29 ± 0.35	4.66 ± 0.49
			310.5	2.23 ± 0.23	3.26 ± 0.34	4.61 ± 0.48
			308.6	2.21 ± 0.23	3.28 ± 0.34	4.63 ± 0.49
			308.2	2.21 ± 0.23	3.28 ± 0.34	4.64 ± 0.49
			308.4	2.21 ± 0.23	3.28 ± 0.34	4.64 ± 0.49
			306.9	2.20 ± 0.23	3.30 ± 0.35	4.66 ± 0.49
			308.3 ± 1.4	2.21 ± 0.23	3.28 ± 0.34	4.64 ± 0.49
119.6	1.121	5.603	255.4	2.24 ± 0.23	3.24 ± 0.34	4.58 ± 0.48
			253.7	2.23 ± 0.23	3.26 ± 0.34	4.61 ± 0.48
			254.4	2.23 ± 0.23	3.25 ± 0.34	4.60 ± 0.48
			251.6	2.21 ± 0.23	3.29 ± 0.34	4.65 ± 0.49
			252.8	2.22 ± 0.23	3.27 ± 0.34	4.63 ± 0.48
			253.5	2.22 ± 0.23	3.26 ± 0.34	4.61 ± 0.48
			253.6 ± 1.4	2.22 ± 0.23	3.26 ± 0.34	4.61 ± 0.48
179.3	0.916	6.860	216.3	2.32 ± 0.24	3.12 ± 0.33	4.42 ± 0.46
			219.4	2.36 ± 0.25	3.08 ± 0.32	4.35 ± 0.46
			218.9	2.35 ± 0.25	3.09 ± 0.32	4.36 ± 0.46
			215.2	2.31 ± 0.24	3.14 ± 0.33	4.44 ± 0.46
			217.8	2.34 ± 0.25	3.10 ± 0.32	4.39 ± 0.46
			217.1	2.33 ± 0.24	3.11 ± 0.33	4.40 ± 0.46
			217.4 ± 1.6	2.34 ± 0.25	3.11 ± 0.33	4.39 ± 0.46
198.7	0.870	7.222	211.6	2.39 ± 0.25	3.03 ± 0.32	4.29 ± 0.45
			213.6	2.42 ± 0.25	3.00 ± 0.31	4.25 ± 0.44
			211.7	2.39 ± 0.25	3.03 ± 0.32	4.29 ± 0.45
			210.0	2.37 ± 0.25	3.06 ± 0.32	4.32 ± 0.45
			210.9	2.38 ± 0.25	3.04 ± 0.32	4.30 ± 0.45
			209.0	2.36 ± 0.25	3.07 ± 0.32	4.34 ± 0.45
			211.1 ± 1.7	2.39 ± 0.25	3.04 ± 0.32	4.30 ± 0.45

3.4 The reconstruction of the Au(111) surface

The reconstruction of the Au(111) surface was investigated by measuring the LEED patterns with lower electron energies. The LEED patterns of the Au(111) surface at two different electron energies are shown in Figure 6. The blue circles indicate the positions of the substrate spots, while the red circles indicate the positions of the reconstruction spots.

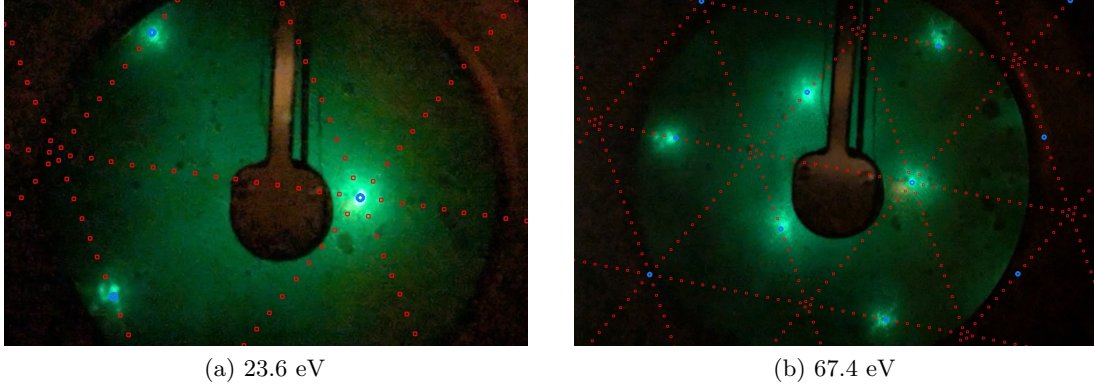


Figure 6: LEED patterns of the Au(111) surface at two different electron energies with the sample rotated showing the characteristic Au(111) reconstruction. The blue circles indicate the positions of the substrate spots. The reconstruction is indicated by the additional red circles in the pattern.

The reconstruction of the Au(111) surface can be observed in the LEED patterns as additional diffraction spots that are not observed by unreconstructed surfaces. The additional diffraction spots are located at positions that are not consistent with the expected positions of the diffraction spots for a perfect fcc surface.

The $(22\times\sqrt{3})$ Au(111) surface reconstruction is a well-known phenomenon that occurs due to the relaxation of the surface atoms. The surface atoms are displaced from their ideal positions, resulting in a distortion of the surface lattice. This distortion leads to the formation of additional diffraction spots in the LEED pattern.^[3]

The reconstruction of the Au(111) surface can be described by a $(22\times\sqrt{3})$ superstructure, which is represented by the following superstructure matrix $\mathbf{M}$:

$$\mathbf{M} = \begin{pmatrix} 11 & 0 \\ 0.5 & 1 \end{pmatrix}.$$

This superstructure matrix $\mathbf{M}$ is used to describe the positions of the additional diffraction spots in the LEED pattern. The superstructure matrix is derived from the reciprocal lattice vectors of the Au(111) surface and the reconstruction.

3.5 PTCDA on the Au(111) surface by epitaxial growth

The growth of PTCDA on the Au(111) surface was investigated using LEED. The LEED pattern of PTCDA on the Au(111) surface shows a superstructure, indicating a long-range ordered arrangement of the molecules on the surface. The LEED pattern of PTCDA on the Au(111) surface at an electron energy of 23.4 eV is shown in Figure 7. The red circles indicate the positions of the center and the purple spots indicate the diffraction spots of the superstructure.

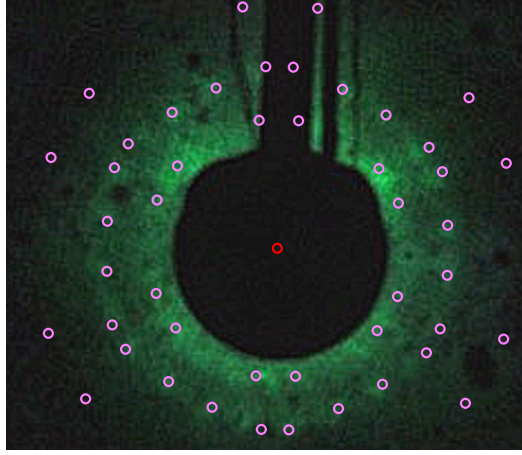


Figure 7: LEED pattern of PTCDA on the Au(111) surface at an electron energy of 23.4 eV. The red circles indicate the positions of the center and the purple spots indicates the diffraction spots of the superstructure.

The LEED pattern of PTCDA on the Au(111) surface shows a superstructure, which is consistent with the expected structure of PTCDA on the Au(111) surface. The superstructure is characterized by a complex unit cell with different rotational domains, which is indicated by the additional diffraction spots in the LEED pattern. The superstructure matrix $\mathbf{M}$ for the PTCDA superstructure is given by:

$$\mathbf{M} = \begin{pmatrix} -0.92 & 4.66 \\ -7.82 & -2.93 \end{pmatrix}.$$

The superstructure matrix $\mathbf{M}$ is used to describe the positions of the additional diffraction spots in the LEED pattern. In comparison with the literature,^[6,7] the superstructure matrix $\mathbf{M}$ is consistent with the expected structure of PTCDA on the Au(111) surface.

4 Discussion

4.1 Determination of the lattice constant of the Au(111) surface

The lattice constant of the Au(111) surface was determined from the LEED pattern by analyzing the positions of the diffraction spots and the distance between the sample and the LEED screen.

The obtained values for the surface lattice constant $\mathbf{a}$ and the bulk lattice constant $\mathbf{a_b}$ of the Au(111) surface are: $\mathbf{a} = 3.17 \pm 0.33 \text{ \AA}$, and $\mathbf{a_b} = 4.49 \pm 0.47 \text{ \AA}$.

The values are in agreement with the literature,^[4] which reports a surface lattice constant of $2.8810 \text{ \AA}$ and a bulk lattice constant of $4.0743 \text{ \AA}$. The difference between the obtained values and the literature values can be attributed to systematic errors in the measurements of distances, such as the determination of the distance between the sample and the LEED screen and the distance of the diffraction spots to the center.

4.2 Reconstruction of the Au(111) surface

The reconstruction of the Au(111) surface is described by a $(22 \times \sqrt{3})$ superstructure. The reconstruction is characterized by the formation of additional diffraction spots in the LEED pattern, which are not observed for unreconstructed surfaces.

The LEED patterns of the Au(111) surface at different electron energies show the characteristic diffraction pattern of the $(22 \times \sqrt{3})$ reconstruction. Furthermore, the additional diffraction spots could be fitted with a superstructure matrix $\mathbf{M}$, which describes the positions of the additional diffraction spots in the LEED pattern.

Nevertheless, the superstructure unit cell used is $(11 \times \frac{\sqrt{3}}{2})$, which leads to a smaller unit cell than the $(22 \times \sqrt{3})$ reconstruction. This could be explained by the fact that the second order spots are higher in intensity than the first order spots and therefore the second order spots are used to fit the superstructure matrix $\mathbf{M}$. This intensity difference with higher intensity for the second order spots could be explained by the fact that the second order spots have strong Fourier coefficients, as stated by Harten *et al.*^[8]

4.3 PTCDA on the Au(111) surface

The LEED pattern of PTCDA on the Au(111) surface shows a superstructure, which is consistent with the expected structure of PTCDA on the Au(111) surface.^[6,7] The superstructure is characterized by a complex unit cell with different rotational domains, which is indicated by the additional diffraction spots in the LEED pattern.

The superstructure matrix $\mathbf{M}$ for the PTCDA superstructure is given by:

$$\mathbf{M} = \begin{pmatrix} -0.92 & 4.66 \\ -7.82 & -2.93 \end{pmatrix}$$

This superstructure matrix $\mathbf{M}$ corresponds to an possible superstructure matrix of the structure $\mathbf{B}_1$, which corresponds to an equilibrium structure of PTCDA on the Au(111) surface.^[6,7] The fact that it is an equilibrium structure is supported by the fact that the structure of the LEED pattern was taken on a sample that was prepared two days before the picture was taken.

5 Appendix

Acronyms

LEED low energy electron diffraction

LEEM low energy electron microscopy

STM scanning tunneling microscopy

XPS x-ray photoelectron spectroscopy

AES auger electron spectroscopy

SPA-LEED spot-profile analysis low energy electron diffraction

PTCDA perylenetetracarboxylic dianhydride

UHV ultra-high vacuum

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